YEAR 8 · SCIENCE DETECTIVES 科学侦探 · WEEK 8 OF 10 · 40 MINUTES

Lab day 检测日

Doing a planned test carefully, repeating it and recording every result straight away (surprises too) makes a result trustworthy evidence.

- I can carry out our plan safely, repeat the test and record results with units in a table.
- I can say what our result matches, how reliable it is and what it does or does not show.

repeat 重复 · reliable 可靠的 · anomaly 异常值 · record 记录 · results table 结果表格 · match 匹配 · settle 沉降 · absorb 吸收 · solvent 溶剂 · mixture 混合物

THE LESSON

Do Now	3	Safety Officer reads kit card aloud; group checks tray against kit request; roles confirmed.
Hook	2	Jury's three questions: repeated? references tested the same way? contaminated?
Learn	3	Safety slide, goggles on; record straight away, units, one line through, circle anomalies.
Do	24	Pairs run parallel repeats, scene sample last; record on sheet; photos 18 to 21; tidy from 21.
Think and Reflect	8	Worksheet; Reporter round, 20 seconds per group; evidence log row 8; exit ticket at the door.

Hands-on: Groups run their Week 7 plan (Soil, Paper, Ink 2 or Powder 2) as two pairs doing parallel repeats.

Collect or assess: Recording sheet and worksheet (strand 3, part of strand 4) into case files; evidence log row 8; exit ticket; photos.

SAFETY

- Goggles for vinegar, iodine, shaking soil bottles; lids tight; plastic bottles only
- Never taste; Powder 2 allergy check (wheat, corn): an allergic student is Recorder
- Soil: cover cuts or wear gloves, wash hands, never down the sink; iodine by dropper

WATCH OUT

- Check every tray before students arrive: with 24 Do minutes, a missing item costs a group its repeat.
- No rubbing out: one line through, circle anomalies. A match is weak or supporting evidence, never proof.

Language: Frame: 'The scene sample matches ___ because ___.' Traps: repeat vs replace, reliable vs accurate. Chinese welcome.

Next week: Week 9: soil bottles stay labelled on the windowsill (look, do not open); keep photos, board notes, strips.

PREP FOR ONE CLASS

before the lesson about 205 min (115 could be done by someone else); after about 78 min (some times estimated)

2 OR MORE WEEKS BEFORE

- [ ] Print job 1: worksheets (48 sheets), run sheet A4 and A3, backup cards
  5 min · reprographics
- [ ] Buy or request consumables; ask canteen for napkins; check filters, tape
  15 min · teacher
- [ ] Collect, rinse 20 identical 300 ml plastic screw-lid bottles per Soil class
  15 min · technician

THE WEEK BEFORE

- [ ] Once options are allocated, print recording sheets x10 and kit cards x10
  5 min · reprographics
- [ ] Write a tray list per group from Week 7 plans; note each class's split
  10 min · teacher
- [ ] Gather prep-room kit, check pens A to D; bag soils A to D and Evidence 6
  50 min · technician
- [ ] Soil: if very wet, dry it on newspaper overnight (optional) optional
  5 min · technician
- [ ] Ink 2 pen pre-test (salt water, vinegar); keep dried strips as answer key
  30 min · teacher

THE DAY BEFORE

- [ ] Powder 2 pots (allergy check first); Ink 2 strips and vinegar; backup cards
  35 min · teacher
- [ ] Paper: make Evidence 7, cut and label P1 to P4, colour the water
  15 min · technician
- [ ] Paper: pre-cut strips if the class is slow with scissors (optional) optional
  10 min · technician

ON THE DAY, BEFORE THE LESSON

- [ ] Fill Soil pots, set out bottles and one checked tray per group
  25 min · technician

AFTER THE LESSON

- [ ] Wash, restock trays; stick in Ink 2 strips; empty soil bottles after Week 9
  30 min · technician
- [ ] Save photos, board notes; read and file work; photograph soil bottles again 45 min · teacher

PRINT

- Worksheet x24 (3 pages on 2 sheets: pages 1 and 2 double-sided)
- Recording sheets x10 (1 sheet, double-sided; 12 if before allocation)
- Kit cards x10 (A5, or 2 per A4 sheet; 4 are for the teacher's file)
- Run sheet A3 enlargement x1 (wall)
- Teacher copies: run sheet A4, backup cards (1 sheet, cut into 4)

GET OR MAKE (NEW THIS LESSON)

- Identical 300 ml plastic screw-lid bottles x20
- Labels x39, markers x2, A4 paper x4, soil bags x5
- Filter strips 2 cm by 10 cm x50, plus spares
- Small plastic cups x44; stirring sticks x22
- Timers x5, scissors x4, bucket x1, extra trays x2
- Teaspoons x3, tablespoons x10, flat sticks x2
- Water jugs x4; measuring jugs or cylinders x4
- Small bottles for Ink 2 vinegar x2 (about 60 ml each)
- Gloves, cloths, paper towels; camera or tablet x1
- White vinegar 180 ml, plus a little for the pre-test
- Baking soda 12, salt 7, cornflour 2, flour 2 level tsp
- Tissues x2, canteen napkins x2, food colouring 6 drops
- Make: soils A to D; Evidence 3, 6 and 7 (x2 each)
- Make: P1 to P4 x2, Powder 2 sets x2, N strips x14
- Make: pre-test strips x6, coloured water 600 ml
- Optional: iodine dropper bottles x2, newspaper

Have ready from earlier weeks: Class box, case files, logbooks, Week 7 plan sheets and option cards, Role cards x6 sets; dictionaries, Goggles x20, magnifiers x5, rulers x7, Trays x6, tiles x4, droppers x2, tape x4, Pots x23, spoons x10, black card x2, Pens A to D x2 sets; kitchen roll, Pencils: 24 writing, 20 for strips